Sequential Testing: ROPE vs Bayes Factor vs LOO

Understanding Divergences in Bayesian Decision Making

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1 Introduction

In this specific example, we will simulate a scenario where we collect data sequentially ($N = 20, 50, 100$) and track how three different Bayesian decision metrics behave:

1. **ROPE (Region of Practical Equivalence)**: Does the effect magnitude matter?
2. **Bayes Factor (via Savage-Dickey)**: Is the null hypothesis ($H_0 : \beta = 0$) less likely than the alternative?
3. **LOO (Leave-One-Out CV)**: Does including the parameter improve predictive accuracy?

We will also see how **prior width** affects the Bayes Factor but has less impact on ROPE and LOO (once N is moderate).

2 Part 1: The Simulation Study

2.1 Data Generation

We generate data where there is a **small but real effect** ($d \approx 0.2$). This is the "danger zone" where decision criteria often disagree.

2.2 The Loop

We will loop through sample sizes $N = \{20, 50, 100\}$. At each step, we fit three models:

1. **Null Model** (**H_0**): `y ~ 1`
2. **Wide Prior Model** (**H_1 Wide**): `y ~ condition`, prior `normal(0, 5)`

3. **Narrow Prior Model (H_1 Narrow)**: `y ~ condition`, prior `normal(0, 0.2)`

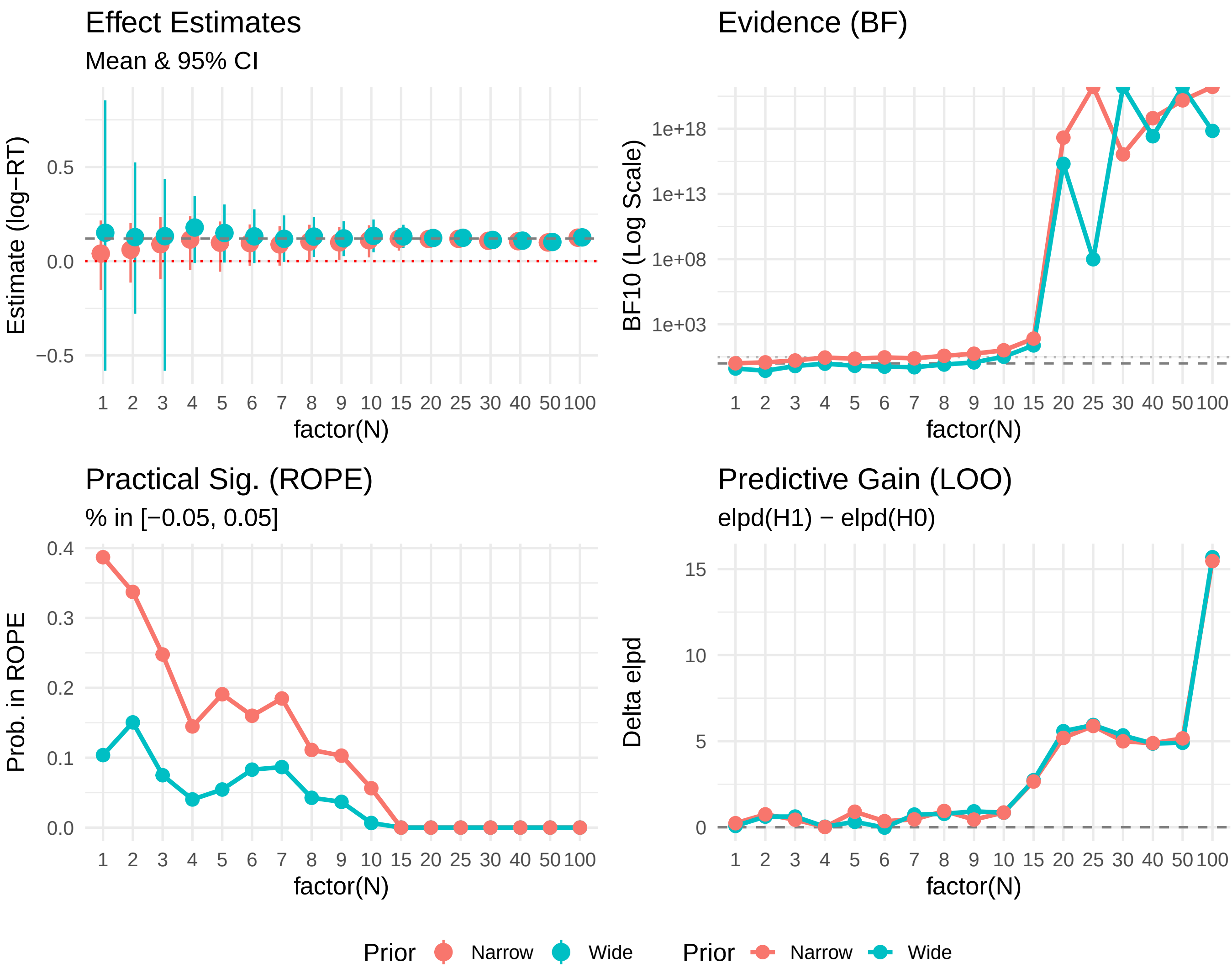
2.3 Results Table

Table 1: Comparison of Decision Metrics by Sample Size and Prior Sensitivity

N	Prior	Estimate_CI	BF10	ROPE_Prob	LOO_Gain
1	Wide	0.15 [-0.58, 0.85]	4.030000e-01	0.104	0.076
1	Narrow	0.04 [-0.15, 0.22]	1.017000e+00	0.387	0.236
2	Wide	0.13 [-0.28, 0.52]	2.720000e-01	0.151	0.620
2	Narrow	0.06 [-0.11, 0.20]	1.195000e+00	0.337	0.750
3	Wide	0.13 [-0.58, 0.44]	6.250000e-01	0.075	0.622
3	Narrow	0.09 [-0.10, 0.23]	1.667000e+00	0.248	0.438
4	Wide	0.18 [-0.01, 0.35]	9.490000e-01	0.040	0.033
4	Narrow	0.11 [-0.05, 0.24]	2.805000e+00	0.145	0.017
5	Wide	0.15 [-0.01, 0.30]	6.530000e-01	0.054	0.319
5	Narrow	0.10 [-0.06, 0.21]	2.303000e+00	0.191	0.904
6	Wide	0.13 [-0.01, 0.28]	5.600000e-01	0.083	-0.018
6	Narrow	0.10 [-0.02, 0.20]	2.872000e+00	0.160	0.354
7	Wide	0.12 [-0.00, 0.24]	5.040000e-01	0.087	0.739
7	Narrow	0.09 [-0.02, 0.19]	2.444000e+00	0.185	0.457
8	Wide	0.13 [0.02, 0.23]	8.170000e-01	0.043	0.779
8	Narrow	0.10 [-0.00, 0.19]	3.818000e+00	0.111	0.946
9	Wide	0.12 [0.03, 0.21]	1.187000e+00	0.037	0.926
9	Narrow	0.10 [0.01, 0.18]	5.508000e+00	0.103	0.447
10	Wide	0.13 [0.05, 0.22]	3.236000e+00	0.007	0.847
10	Narrow	0.11 [0.02, 0.19]	1.012300e+01	0.056	0.857
15	Wide	0.13 [0.07, 0.19]	2.394000e+01	0.000	2.735
15	Narrow	0.12 [0.06, 0.18]	8.004300e+01	0.000	2.657
20	Wide	0.12 [0.08, 0.17]	2.040489e+15	0.000	5.582
20	Narrow	0.12 [0.07, 0.16]	2.093883e+17	0.000	5.193
25	Wide	0.12 [0.08, 0.16]	9.728796e+07	0.000	5.943
25	Narrow	0.12 [0.08, 0.16]	Inf	0.000	5.885
30	Wide	0.11 [0.07, 0.15]	Inf	0.000	5.335
30	Narrow	0.11 [0.07, 0.15]	1.081383e+16	0.000	4.994
40	Wide	0.11 [0.07, 0.14]	2.675363e+17	0.000	4.863
40	Narrow	0.11 [0.07, 0.14]	6.526253e+18	0.000	4.884
50	Wide	0.10 [0.07, 0.13]	Inf	0.000	4.907
50	Narrow	0.10 [0.07, 0.13]	1.511938e+20	0.000	5.163
100	Wide	0.13 [0.10, 0.15]	6.898743e+17	0.000	15.687
100	Narrow	0.12 [0.10, 0.15]	Inf	0.000	15.463

2.4 Visualization of Divergence

Let's visualize how these metrics evolve as N increases.



2.4.1 Interpretation

1. **Bayes Factor Divergence**: Notice that `BF10_Wide` is significantly lower than `BF10_Narrow`. The wide prior penalizes the alternative hypothesis because it spreads probability density over a huge range where data is not found ("Dilution Effect").
2. **Small N (20)**: